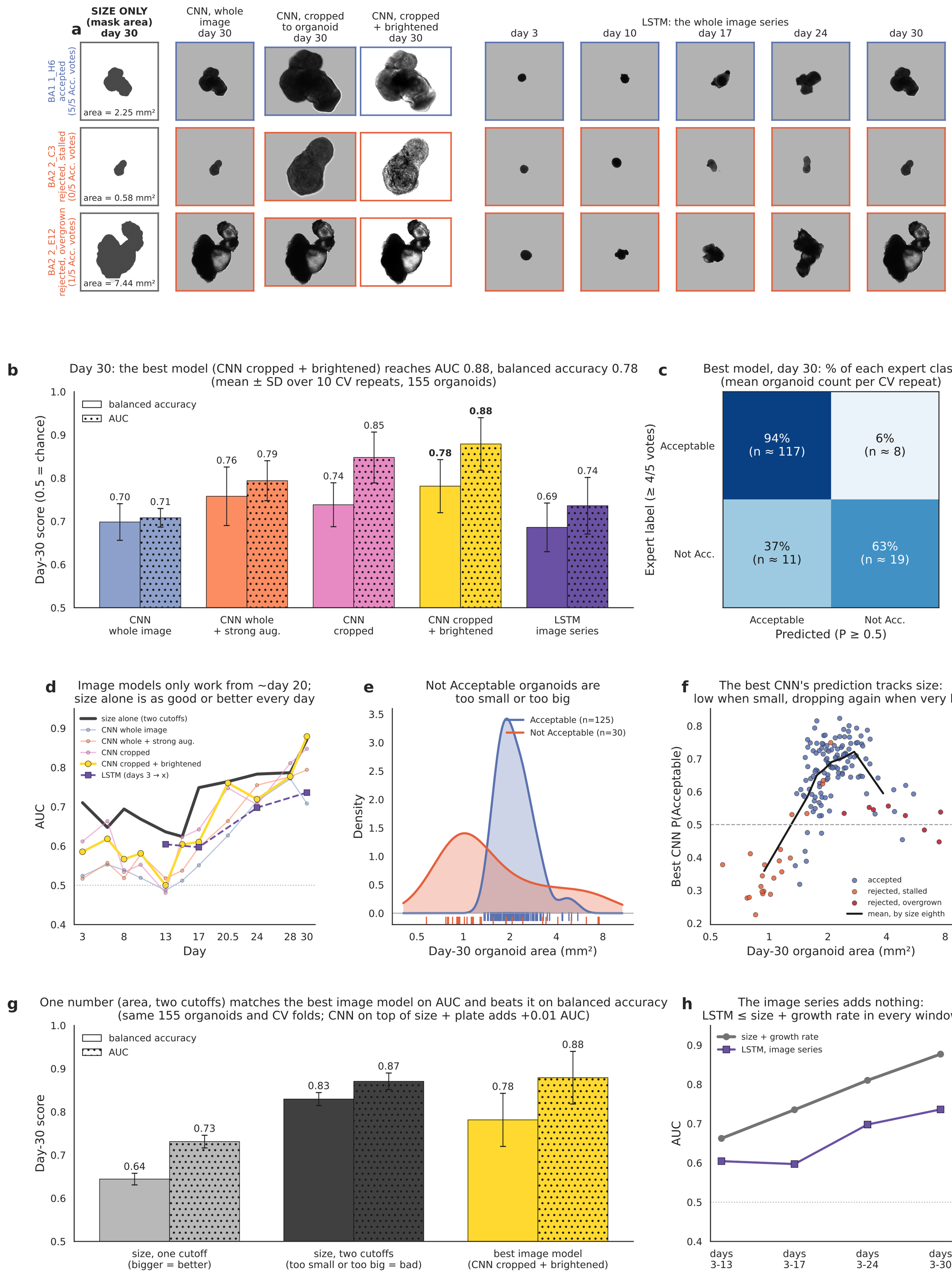

a
SIZE ONLY (mask area) day 30
CNN, whole image day 30
CNN, cropped to organoid day 30
CNN, cropped + brightened day 30
LSTM: the whole image series
day 3
day 10
day 17
day 24
day 30
BA1 1_H6 accepted (5/5 Acc. votes)
area = 2.25 mm²
BA2 2_C3 rejected, stalled (0/5 Acc. votes)
area = 0.58 mm²
BA2 2_E12 rejected, overgrown (1/5 Acc. votes)
area = 7.44 mm²
b
Day 30: the best model (CNN cropped + brightened) reaches AUC 0.88, balanced accuracy 0.78
(mean ± SD over 10 CV repeats, 155 organoids)
balanced accuracy
AUC
Day-30 score (0.5 = chance)
0.70
0.71
0.76
0.79
0.74
0.85
0.78
0.88
0.69
0.74
CNN whole image
CNN whole + strong aug.
CNN cropped
CNN cropped + brightened
LSTM image series
c
Best model, day 30: % of each expert class
(mean organoid count per CV repeat)
Expert label (≥ 4/5 votes)
Acceptable
Not Acc.
94% (n ≈ 117)
6% (n ≈ 8)
37% (n ≈ 11)
63% (n ≈ 19)
Predicted (P ≥ 0.5)
d
Image models only work from ~day 20; size alone is as good or better every day
size alone (two cutoffs)
CNN whole image
CNN whole + strong aug.
CNN cropped
CNN cropped + brightened
LSTM (days 3 → x)
AUC
Day
e
Not Acceptable organoids are too small or too big
Acceptable (n=125)
Not Acceptable (n=30)
Density
Day-30 organoid area (mm²)
f
The best CNN's prediction tracks size: low when small, dropping again when very big
Best CNN P(Acceptable)
accepted
rejected, stalled
rejected, overgrown
mean, by size eighth
Day-30 organoid area (mm²)
g
One number (area, two cutoffs) matches the best image model on AUC and beats it on balanced accuracy
(same 155 organoids and CV folds; CNN on top of size + plate adds +0.01 AUC)
balanced accuracy
AUC
Day-30 score
0.64
0.73
0.83
0.87
0.78
0.88
size, one cutoff (bigger = better)
size, two cutoffs (too small or too big = bad)
best image model (CNN cropped + brightened)
h
The image series adds nothing: LSTM ≤ size + growth rate in every window
size + growth rate
LSTM, image series
AUC
days 3-13
days 3-17
days 3-24
days 3-30